La Scuola d'Italia / Dr. Huson / IB Math 12
9 September 2026

First & Last name:

2.1 Classwork: Exponential Functions Review

1. [Maximum mark: 7]

   Consider the function $f(x) = 3 \cdot 2^x - 5$.

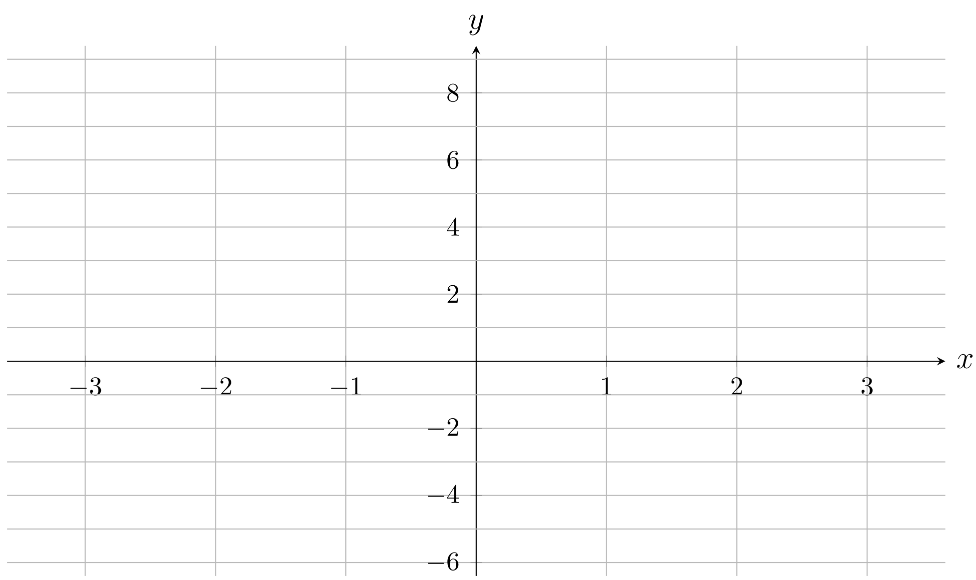
   (a) Write down the equation of the horizontal asymptote of the graph of f. [1]

   (b) Find the y-intercept of the graph of f. [1]

   (c) Find $f(4)$. [1]

   (d) Find the x-intercept of the graph of f, correct to three significant figures. [2]

   (e) On the axes below, sketch the graph of f for $-3 \leq x \leq 2$, showing the horizontal asymptote and both intercepts. [2]



2. [Maximum mark: 6]

Solve each equation for the unknown, showing your working with logarithms. Give each answer correct to three significant figures.

(a) $5^{x} = 300$ [2]

(b) $20\,e^{0.15t} = 90$ [2]

(c) $2^{x+1} = 3^{x}$ [2]

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3. [Maximum mark: 7]

The number of electric cars registered in a city is modelled by

$$N(t) = N_0 e^{kt}, \qquad k > 0,$$

where t is the number of years after the start of 2020. At the start of 2022 there were 4500 electric cars registered, and at the start of 2026 there were 7200.

(a) Find the value of k, correct to three significant figures. [2]

(b) Find the value of N_0, correct to the nearest whole number. [1]

(c) The model predicts that the number of registered electric cars grows by a fixed percentage each year. Find this annual percentage growth rate, correct to three significant figures. [1]

(d) Use the model to predict the number of registered electric cars at the start of 2030, correct to the nearest whole number. [1]

(e) Find the value of t at which the number of registered electric cars first reaches 15 000. Give your answer correct to three significant figures. [2]

4. [Maximum mark: 6]

Ana invests \$2000 in an account paying interest of 4.5 % per annum, compounded annually. The value of her investment after n years is given by the formula

$$FV = PV\left(1+\frac{r}{100}\right)^n,$$

where PV is the amount invested and r is the annual interest rate. No further deposits or withdrawals are made.

(a) Find the value of Ana's investment after 8 years, correct to the nearest cent. [1]

(b) Find the smallest whole number of years after which Ana's investment is worth at least twice the amount she invested. [2]

On the same day, Ben invests \$3000 in an account paying interest of 2 % per annum, compounded annually.

(c) Find the smallest whole number of years after which Ana's investment is worth more than Ben's. [3]